

Intervalos de confianza

Parámetro	Varianza conocida y otras condiciones	Estimador	Límite inferior	Límite superior
μ	Si	$Z = \frac{\bar{X} - \bar{x}}{\sigma/\sqrt{n}}$	$L_i = \bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$	$L_s = \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$
	No $n \geq 30$	$Z = \frac{\bar{X} - \bar{x}}{s/\sqrt{n}}$	$L_i = \bar{x} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$	$L_s = \bar{x} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$
	No $n < 30$ Distribución Normal	$t_{n-1} = \frac{\bar{X} - \bar{x}}{s/\sqrt{n}}$	$L_i = \bar{x} - t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}$	$L_s = \bar{x} + t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}$
p	$n \geq 30$	$Z = \frac{\hat{P} - \hat{p}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$	$L_i = \hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$L_s = \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
σ^2	No Distribución Normal	$\chi^2(n-1)$	$L_i = \frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}^2}$	$L_s = \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}$

Siendo:

$$z_{\frac{\alpha}{2}} = \text{cuantil} \left(1 - \frac{\alpha}{2} \right) \text{ de } Z: N(0,1) \rightarrow P \left\{ Z < z_{\frac{\alpha}{2}} \right\} = 1 - \frac{\alpha}{2} \quad (\text{Tabla A4})$$

$$t_{\frac{\alpha}{2}, n-1} = \text{cuantil} \left(1 - \frac{\alpha}{2} \right) \text{ de } T: \text{Student}(n-1) \rightarrow P \left\{ T < t_{\frac{\alpha}{2}, n-1} \right\} = 1 - \frac{\alpha}{2} \rightarrow P \left\{ T > t_{\frac{\alpha}{2}, n-1} \right\} = \frac{\alpha}{2} \quad (\text{Tabla A5})$$

$$\chi_{\frac{\alpha}{2}, n-1}^2 = \text{cuantil} \left(1 - \frac{\alpha}{2} \right) \text{ de } \chi^2: \text{Chi-cuadrado}(n-1) \rightarrow P \left\{ \chi^2 < \chi_{\frac{\alpha}{2}, n-1}^2 \right\} = 1 - \frac{\alpha}{2} \rightarrow P \left\{ \chi^2 > \chi_{\frac{\alpha}{2}, n-1}^2 \right\} = \frac{\alpha}{2} \quad (\text{Tabla A6})$$

$$\chi_{1-\frac{\alpha}{2}, n-1}^2 = \text{cuantil} \left(\frac{\alpha}{2} \right) \text{ de } \chi^2: \text{Chi-cuadrado}(n-1) \rightarrow P \left\{ \chi^2 < \chi_{1-\frac{\alpha}{2}, n-1}^2 \right\} = \frac{\alpha}{2} \rightarrow P \left\{ \chi^2 > \chi_{1-\frac{\alpha}{2}, n-1}^2 \right\} = 1 - \frac{\alpha}{2} \quad (\text{Tabla A6})$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Intervalos de confianza

Diferencia de parámetros	Varianzas conocidas y otras condiciones	Límite inferior	Límite superior
$\mu_X - \mu_Y$	Si	$L_i = \bar{x} - \bar{y} - z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$	$L_s = \bar{x} - \bar{y} + z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
	No $n, m \geq 30$	$L_i = \bar{x} - \bar{y} - z_{\frac{\alpha}{2}} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$	$L_s = \bar{x} - \bar{y} + z_{\frac{\alpha}{2}} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$
	No $n, m < 30$ $\sigma_X = \sigma_Y$ Distribución Normal	$L_i = \bar{x} - \bar{y} - \left(t_{\frac{\alpha}{2}, n+m-2}\right)(s_p) \sqrt{\frac{1}{n} + \frac{1}{m}}$	$L_s = \bar{x} - \bar{y} + \left(t_{\frac{\alpha}{2}, n+m-2}\right)(s_p) \sqrt{\frac{1}{n} + \frac{1}{m}}$
	No $n, m < 30$ $\sigma_X \neq \sigma_Y$ Distribución Normal	$L_i = \bar{x} - \bar{y} - t_{\frac{\alpha}{2}, v} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$	$L_s = \bar{x} - \bar{y} + t_{\frac{\alpha}{2}, v} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$
	Muestras pareadas $n \geq 30$	$L_i = \bar{d} - z_{\frac{\alpha}{2}} \frac{s_D}{\sqrt{n}}$	$L_s = \bar{d} + z_{\frac{\alpha}{2}} \frac{s_D}{\sqrt{n}}$
	Muestras pareadas $n < 30$ Distribución Normal	$L_i = \bar{d} - t_{\frac{\alpha}{2}, n-1} \frac{s_D}{\sqrt{n}}$	$L_s = \bar{d} + t_{\frac{\alpha}{2}, n-1} \frac{s_D}{\sqrt{n}}$
$p_1 - p_2$	$n, m \geq 30$	$L_i = \hat{p}_1 - \hat{p}_2 - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	$L_s = \hat{p}_1 - \hat{p}_2 + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

Siendo:

$$s_p = \sqrt{\frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2}}$$

$$v = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{s_X^4}{n^2(n-1)} + \frac{s_Y^4}{m^2(m-1)}}$$

$$D = X - Y$$

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s_Y^2 = \frac{1}{m-1} \sum_{i=1}^m (y_i - \bar{y})^2$$